

Intersection excess controls partite cover numbers

Research continuation, 7 October 2026. This develops the same partite cover-number programme beyond its finite $13/4$ and asymptotic $10/3$ bounds. The new result records the intersection structure, not only another numerical bound.

1. Definitions and main theorem

Let H be a finite simple intersecting r -partite r -uniform hypergraph, $r \geq 2$. Its edges contain exactly one vertex in each of the specified r parts. Write $m = |E(H)|$, $t = \tau(H)$, and define its **intersection excess** by

$$I(H) = \sum_{\{A,B\} \subset E(H)} (|A \cap B| - 1) \geq 0. \quad (1)$$

The sum is over unordered pairs of distinct edges. In particular $I(H) = 0$ precisely when every distinct edge pair meets in exactly one vertex. This is the linear intersecting class. For an empty family, both the cover number and the excess are zero.

Put

$$d = \frac{27 - 5\sqrt{17}}{20}, \quad a = 5d = \frac{27 - 5\sqrt{17}}{4}, \quad c = 5 - a = \frac{5\sqrt{17} - 7}{4} > \frac{10}{3}. \quad (2)$$

The strict comparison follows from $15\sqrt{17} > 61$, verified by squaring positive sides: $225 \cdot 17 = 3825 > 3721 = 61^2$.

Theorem 1 (cover number with intersection-excess penalty). For every $\delta > 0$ there is a finite constant C_δ , independent of r, H , such that

$$m \geq 5\tau(H) - (a + \delta)r - \frac{10I(H)}{r} - C_\delta. \quad (3)$$

Equivalently this bounds $\tau(H)$ by $m/5 + (d + \delta/5)r + 2I(H)/r + C_\delta/5$.

Corollary 2 (linear and near-linear families). Suppose $r_j \rightarrow \infty$, H_j satisfies the hypotheses, $\tau(H_j) \geq r_j - 1$, and $I(H_j) = o(r_j^2)$. Then

$$\liminf_j \frac{|E(H_j)|}{r_j} \geq c = \frac{5\sqrt{17} - 7}{4} = 3.403882\dots \quad (4)$$

In particular, the minimum edge count $f_{\text{lin}}(r)$ under the additional linearity hypothesis satisfies $f_{\text{lin}}(r) \geq (c - \varepsilon)r$ for every fixed $\varepsilon > 0$ and all sufficiently large ranks at which its defining class is nonempty.

Corollary 3 (quadratic excess required near the $10/3$ coefficient). If instead $|E(H_j)|/r_j \rightarrow 10/3$ and $\tau(H_j) \geq r_j - 1$, then

$$\liminf_j \frac{I(H_j)}{r_j^2} \geq \frac{15\sqrt{17} - 61}{120} > 0. \quad (5)$$

This is a necessary lower bound supplied by the proof; optimality or existence of such sequences is not asserted.

These statements do not solve Ryser's conjecture. The class $\tau \geq r - 1$ includes both equality examples and hypothetical $\tau = r$ counterexamples. No existence for every rank, numerical asymptotic threshold or priority is claimed. The finite literature screen is recorded separately. The new step here is the simultaneous weighted linearization of degree-three and degree-four blocks, and its coupling to the part cover and a degree-four star bound. The peeling, pair-excess deletion framework and matching/star tools have explicit predecessors in Sivashankar [S, Section 4].

2. Attributed inputs

We use Kahn's small-codegree edge-colouring theorem in the form stated in [KKKMO, Theorem 3.1]. For every $\eta > 0$ there is an integer $D_0 \geq 1$ such that every linear hypergraph of rank at most four and maximum degree $D \geq D_0$ admits a proper edge-colouring with at most $(1 + \eta)D$ colours. Here rank is maximum edge size; linearity means codegree at most one. This follows by fixing rank four in Kahn's theorem and choosing D_0 so that its small-codegree condition includes codegree one. It applies both to four-uniform and mixed three/four-uniform hypergraphs used below. We do not re-prove this standard theorem or formalize it in Lean.

We also use Sivashankar's degree-three lemma: if L is intersecting and r -uniform, has h edges and maximum degree at most three, then $4\tau(L) \leq h + r + 4$. Its full attributed proof is included in the appendix. No OpenAI/math theorem is used.

3. Residual notation and exact counts

Peel vertices of current degree at least five, recording each selected vertex and deleting all its current incident edges. Stop at a residual J of maximum degree at most four. If k vertices were selected and $q = |E(J)|$, then

$$m \geq q + 5k, \quad t \leq k + \tau(J), \quad m \geq q + 5(t - \tau(J)). \quad (6)$$

Fixing one residual edge and counting its intersections gives $q \leq 3r + 1$; this also holds for an empty residual. Thus

$$q \leq 7r/2 \quad (r \geq 2). \quad (7)$$

The intersection excess of the residual is $T = I(J) \leq I(H)$, because the sum omits some nonnegative summands and intersections themselves do not change under edge deletion.

Write x_i for its degree- i vertex counts, and put $W = x_3/2 + x_4$, $K = q(q - r)/2$. Incidence and pair counting give

$$x_1 + 2x_2 + 3x_3 + 4x_4 = qr, \quad T = x_2 + 3x_3 + 6x_4 - \binom{q}{2}.$$

Consequently

$$x_4 + 3W = K - q/2 + T + x_1/2 \geq K - q/2 + T. \quad (8)$$

Set $s = q/2 + 1 - \tau(J)$. Pairing residual edges proves $s \geq 0$. Selecting the degree-three/four vertices of a single part, followed by pairing the remaining edges, saves $a_c/2 + b_c$ in that part. Their sum across parts is W , hence

$$W \leq rs. \quad (9)$$

This is the same part-cover argument as in the preceding notes: the incident edge sets are disjoint within a part, and $\lceil n/2 \rceil \leq n/2 + 1/2$ for integer $n \geq 0$.

4. Degree-four linearization and its uniform bound

View the edges of J as points. A residual vertex defines its incidence block. In the multihypergraph of degree-four blocks, pair codegrees are bounded by the original intersection sizes. Therefore their total positive pair excess is at most T . Delete a block containing a pair of codegree at least two until all codegrees are at most one. Each deletion lowers that integer excess by at least one; at most T blocks are deleted. The retained simple linear four-uniform hypergraph has at least $x_4 - T$ blocks.

For a retained graph of maximum degree $D \geq D_0$, Kahn's colouring gives a matching of at least $e/((1+\eta)D)$ blocks, so $s \geq e/((1+\eta)D)$. A star through a maximum-degree point has exactly $1+3D$ points by linearity. Its defining vertices and a pairing of the rest give $s \geq D/2$. Thus $e \leq 2(1+\eta)s^2$. For $D < D_0$, incidence counting instead gives $4e \leq qD_0$. These include empty retained graphs. In either case

$$x_4 \leq T + 2(1+\eta)s^2 + qD_0/4. \quad (10)$$

Combining (8)–(10), with $B_0 = 1/2 + D_0/4$, yields

$$K \leq 2(1+\eta)s^2 + 3rs + qB_0. \quad (11)$$

The T terms cancel in this first inequality.

5. Weighted three/four-block linearization

Now form the mixed multihypergraph of all degree-three and degree-four incidence blocks. Give the former weight $1/2$ and the latter weight one. The total weight is W . Its pair excess is at most T , again because its pair codegrees are at most $|A \cap B|$. The same deletion procedure produces a simple linear rank-four hypergraph G , whose edges have sizes three or four. Each deletion loses weight at most one, so its total retained weight U satisfies

$$U \geq W - T. \quad (12)$$

The right-hand side need not be positive. Let D be the maximum degree of G . Through one point, all other points in its incident blocks are disjoint, and each block supplies at least two such points. Therefore $2D \leq q - 1$ whenever G is nonempty.

If $D \geq D_0$, a proper colouring with at most $(1+\eta)D$ colours has a colour class of weight at least $U/((1+\eta)D)$. It is a matching. If it contains a_3 three-blocks and a_4 four-blocks, its defining vertices and a pairing cover give

$$\tau(J) \leq a_3 + a_4 + \lceil (q - 3a_3 - 4a_4)/2 \rceil \leq q/2 + 1/2 - a_3/2 - a_4.$$

Hence s is at least that colour-class weight, and $U \leq (1+\eta)Ds \leq (1+\eta)qs/2$. If $D < D_0$, total weight is at most the number of blocks, and each block has at least three points. Thus $U \leq qD_0/3$. Both cases, including empty G , give

$$W \leq T + (1+\eta)qs/2 + qD_0/3. \quad (13)$$

Use (8), (10) and (13) to obtain the second inequality

$$K \leq 2(1+\eta)s^2 + \frac{3}{2}(1+\eta)qs + 3T + qB_1, \quad B_1 = 1/2 + 5D_0/4. \quad (14)$$

6. A piecewise scalar certificate and the defect penalty

The constant d in (2) obeys

$$3/10 \leq d \leq 1/3, \quad d^2 - 27d/10 + 19/25 = 0, \quad d^2 \leq 1/9 < 13/100. \quad (15)$$

For the lower interval use $\sqrt{17} \leq 21/5$; for the upper interval use $\sqrt{17} \geq 61/15$; both follow by squaring positive sides. Substitution verifies the quadratic identity.

If $q \leq r$, the inherited degree-three lemma, applied after a maximal matching of four-blocks, gives $\tau(J) \leq (q+r+4)/4$. Equation (6) then yields $m \geq 5t - 5r/4 - q/4 - 5 \geq 5t - 3r/2 - 5$. As $5d \geq 3/2$, this is stronger than (3). Thus assume $q \geq r$.

Set

$$b = (1 + \eta)s, \quad \kappa = \min\{r, q/2\}, \quad L = 3T + qB_1, \quad E = 2L/(3r), \quad \ell = 3q/10 - dr.$$

Since $b \geq s \geq 0$, (11) and (14) imply

$$2b^2 + 3\kappa b \geq K - L. \quad (16)$$

Also $\kappa \geq r/2$ and $E \geq 0$. Two exact factorizations, using the quadratic identity in (15), prove

$$K \geq 2\ell^2 + 3\kappa\ell. \quad (17)$$

For $r \leq q \leq 2r$, the difference equals

$$(2r - q)(13q/100 - d^2r) \geq 0.$$

For $q \geq 2r$, it equals

$$(q - 2r)(8q/25 + (6d/5 - 19/25)r) \geq 0.$$

The second factor here is at least $(6d/5 - 3/25)r \geq 6r/25$. The first factorization uses $q \geq r$ and $d^2 < 13/100$.

It follows that $b + E \geq \ell$. Otherwise $\ell - b > E$, with $\ell > b \geq 0$, and

$$(2\ell^2 + 3\kappa\ell) - (2b^2 + 3\kappa b) = (\ell - b)(2(\ell + b) + 3\kappa) > 3\kappa E \geq (3r/2)E = L,$$

contradicting (16)–(17). This also handles a negative ℓ , when the asserted lower bound is immediate.

By (7), $\ell \leq r$ and $E \leq 2T/r + 7B_1/3$. If $s \leq r$, the inequality $(1 + \eta)s + E \geq \ell$ implies $s \geq \ell - \eta r - E$; if $s > r$, that conclusion follows from $\ell \leq r$ and $E \geq 0$. Hence (6) gives

$$\begin{aligned} m &\geq 5t - 3q/2 + 5s - 5 \\ &\geq 5t - (5d + 5\eta)r - 10T/r - 35B_1/3 - 5 \\ &\geq 5t - (5d + 5\eta)r - 10I(H)/r - 35B_1/3 - 5. \end{aligned} \quad (18)$$

Set $\eta = \delta/5$. This proves Theorem 1 with the finite, possibly ineffective constant

$$C_\delta = 5 + \frac{35}{3} \left(\frac{1}{2} + \frac{5D_0(\delta/5)}{4} \right). \quad (19)$$

An empty original hypergraph satisfies the result immediately.

For Corollary 2, divide (3) by r_j , use $5\tau(H_j)/r_j \geq 5 - 5/r_j$, take the lower limit, and let $\delta \downarrow 0$. For Corollary 3, the same inequality gives

$$\frac{10I(H_j)}{r_j^2} \geq c - \delta - \frac{|E(H_j)|}{r_j} - \frac{C_\delta + 5}{r_j}.$$

Take the lower limit and then $\delta \downarrow 0$, noting $(c - 10/3)/10 = (15\sqrt{17} - 61)/120$. $\square$

7. Verification and contribution limits

The new global excess penalty follows from weighted block matching, not from a finite search. The exact checker replays both block deletions and weighted matching covers, and checks the quadratic-field identities without floating point. Lean formalizes scalar identities and inequalities under their explicit hypotheses. The finite combinatorics, Kahn’s theorem and the coloured-graph classification in the appendix remain written arguments. No complete Lean formalization or external human review is claimed. Priority has not been determined.

The result separates possible near-minimal families from the linear regime through an explicit intersection-excess obstruction. It does not classify the nonlinear configurations or prove the constants optimal. Those are the remaining mathematical gaps.

References

- [S] V. Sivashankar, *An Improved Lower Bound for the Erdős–Lovász Cover Number Problem*, arXiv:2606.24878v2. The degree-three lemma is Lemma 2; Section 4 supplies the existing peeling, pair-excess linearization and matching/star framework. The present weighted three/four-block estimate and its global defect consequence are distinguished from that framework.
- [K] J. Kahn, *Asymptotically good list-colorings*, Journal of Combinatorial Theory, Series A 73 (1996), 1–59.
- [KKKMO] D. Y. Kang, T. Kelly, D. Kühn, A. Methuku and D. Osthus, *Solution to a problem of Erdős on the chromatic index of hypergraphs with bounded codegree*, Proceedings of the London Mathematical Society 129 (2024), no. 6, e70011, arXiv:2110.06181, Theorem 3.1. Its rank bound covers mixed three/four-block hypergraphs.
- [ABW] R. Aharoni, J. Barát and I. M. Wanless, *Multipartite hypergraphs achieving equality in Ryser’s conjecture*, Graphs and Combinatorics 32 (2016), 1–15, arXiv:1409.4833v2, DOI:10.1007/s00373-015-1575-9. The directly applicable general coefficient there is 293/96.

Appendix. Full proof of the inherited degree-three lemma

Lemma A (Sivashankar [S, Lemma 2]). For an intersecting r -uniform hypergraph L with q edges and maximum degree at most three,

$$4\tau(L) \leq q + r + 4. \quad (\text{A.1})$$

The proof below restates its maximum-matching argument in our notation; this lemma and its local coloured-graph claim are not claimed as our contribution.

Proof. The empty case is immediate. Fixing one edge and counting its intersections with all other edges gives

$$q - 1 \leq \sum_{v \in e} (d(v) - 1) \leq 2r, \quad q \leq 2r + 1. \quad (\text{A.2})$$

Treat the q edges as points. Each original vertex defines the block of points containing it, of size at most three. Every point pair is contained in a block. Take a maximum matching of the distinct three-point blocks; let its size be s , its covered points C , and its unmatched points U , with $|U| = u$. Thus $q = 3s + u$. There is no three-block inside U . The defining vertices of the matching, followed by a pairing of U , give a cover of size at most $s + \lceil u/2 \rceil$. It suffices to show

$$r \geq u + s - 2. \quad (\text{A.3})$$

Put $\alpha = r - u - s + 2$, an integer. If $u = 0$, (A.2) gives $\alpha \geq (r + 5)/3 > 0$. Assume $u > 0$.

Select one witness vertex for each pair in U . These witnesses are distinct: witnesses for disjoint pairs cannot coincide because the maximum degree is three; witnesses for overlapping pairs cannot coincide because that would create a three-block inside U . Each selected witness contributes exactly two incidences with U ; its block is the selected pair, possibly together with one point of C . The witnesses contribute $u(u - 1)$ such incidences in total.

We need the following coloured-graph observation, setting $K = \max\{12, u + 4\}$. In a simple graph on u vertices, colour each edge with one of at most three colours. If two disjoint edges never have different colours, the sum of the three colour support sizes is at most K . Here a colour support consists of the vertices incident with edges of that colour.

To prove the observation, suppose first that colour one contains two disjoint edges e, f , and put $W = e \cup f$. Every edge of the other colours must meet both, so its endpoints belong to W . If no other colour is used, the assertion holds. If their combined edge family has no common vertex, each colour-one edge must also have both endpoints in W : an edge with an endpoint outside W would force its other endpoint to be common to that family. All three support sizes are then at most four, giving twelve. Otherwise the combined family has a common vertex $c \in W$. Its edges must join c to one of the two endpoints of the edge among e, f not containing c . There are at most two such edges, and simplicity makes their combined colour-support sum at most four. Adding the first support gives $u + 4$.

Now assume each colour class is pairwise intersecting. A simple pairwise-intersecting graph is a star or is contained in a triangle: take two edges ab, ac ; any edge avoiding a must be bc , which in turn forces every edge into that triangle. If all supports have size at most four, their sum is at most twelve. Otherwise one class is a star with at least four leaves. Every edge of the other colours must contain its centre, since two endpoints cannot meet four distinct leaves. Simplicity ensures each noncentral vertex belongs to at most one support. The centre belongs to at most three, giving a sum at most $u + 2$. This proves the observation.

For each matching triple M_i , form a coloured graph on U : the selected pair xy is given colour $z \in M_i$ when its witness block is $\{x, y, z\}$. Two disjoint pairs with different colours would allow replacing M_i by two disjoint three-blocks, increasing the matching. Thus the observation applies. The colour support sum counts the distinct pairs between U and M_i covered by selected witnesses, so at most K such pairs are covered. Across the s triples, at least $3su - sK$ of the U - C pairs are uncovered by selected witnesses.

For a nonselected vertex let a count its block points in U and c those in C . Since $a + c \leq 3$, one has $ac \leq 2a$. All remaining pairs have nonselected witnesses, so their incidence contribution to U is at least $(3su - sK)/2$. Counting the total ur incidences with U yields

$$ur \geq u(u - 1) + \frac{3su - sK}{2}. \quad (\text{A.4})$$

This reasoning remains valid if the displayed lower bound is negative.

Equation (A.2) with $r = u + s - 2 + \alpha$ gives $s \leq u - 3 + 2\alpha$. Dividing (A.4) by $u > 0$ gives

$$\alpha \geq 1 - \frac{(K - u)s}{2u}. \quad (\text{A.5})$$

If $\alpha < 0$, its integrality implies $\alpha \leq -1$ and $0 \leq s \leq u - 5$, hence $u \geq 5$. As $K - u \geq 0$, (A.5) implies $\alpha \geq 1 - (K - u)(u - 5)/(2u)$. This is greater than -1 . Indeed, for $5 \leq u \leq 8$ one has $K = 12$ and

$$4u - (12 - u)(u - 5) = u^2 - 13u + 60 = (u - 13/2)^2 + 71/4 > 0.$$

For $u \geq 8$, one has $K = u + 4$, giving $-1 + 10/u > -1$. This contradiction proves $\alpha \geq 0$, hence (A.3).
 Finally

$$4\tau(L) \leq 4s + 4\lceil u/2 \rceil \leq 4s + 2u + 2 \leq q + r + 4.$$

The lemma follows. $\square$